# Llama Index vs LangChain

Wednesday, July 29, 2026 7:55 PM

## Agenda:

1. What is llama Index?
2. What is Lang Chain?
3. What is difference between Llama Index and Lang Chain?
4. Where we should which one?
5. Can we combine both for our application building?

## What is Llama Index?

Llama Index is an AI data framework that helps Large Language Models (LLMs) like GPT, Gemini, Llama, Mistral, etc. connect to your own data.

Think of it as a bridge between your data and an LLM.

**Layman Analogy:**
**Imagine:**

- LLM (ChatGPT/Gemini) = A very intelligence student.
- Your documents = Thousands of books in a library.
- LlamaIndex = The librarian.

## Llama Index supported LLMs:

OpenAI (GPT), Google Gemini, Anthropic Claude, Meta Llama, Mistral, Cohere, Ollama (Local models), Hugging Face models, NVIDIA NIMs.

**Major Components of Llama Index:**

| Component | Purpose |
|---|---|
| Readers | Load data from different sources |
| Documents | Store loaded text |
| Nodes | Small chunks of documents |
| Embeddings | Convert text into vectors |
| Index | Organize data for retrieval |
| Retriever | Find relevant chunks |
| Query Engine | Answer user questions |
| Chat Engine | Enable conversational interactions |
| Response Synthesizer | Combine retrieved context into a final answer |

## What is LangChain?

**LangChain** is an **open-source framework** for building applications powered by **Large Language Models (LLMs)**.

If **LlamaIndex** is mainly responsible for **connecting LLMs to your data**, then **LangChain** is responsible for **orchestrating the entire AI application's workflow**.

**Layman Analogy**
Imagine you're building an AI travel assistant.

- **LLM (GPT/Gemini)** = A smart travel expert.
- **LlamaIndex** = The librarian who finds travel information.
- **LangChain** = The travel manager who coordinates everything.

The travel manager can:

- Ask the librarian for information.
- Check flight prices using an API.
- Book a hotel.
- Remember previous conversations.
- Calculate travel costs.
- Return one final response.

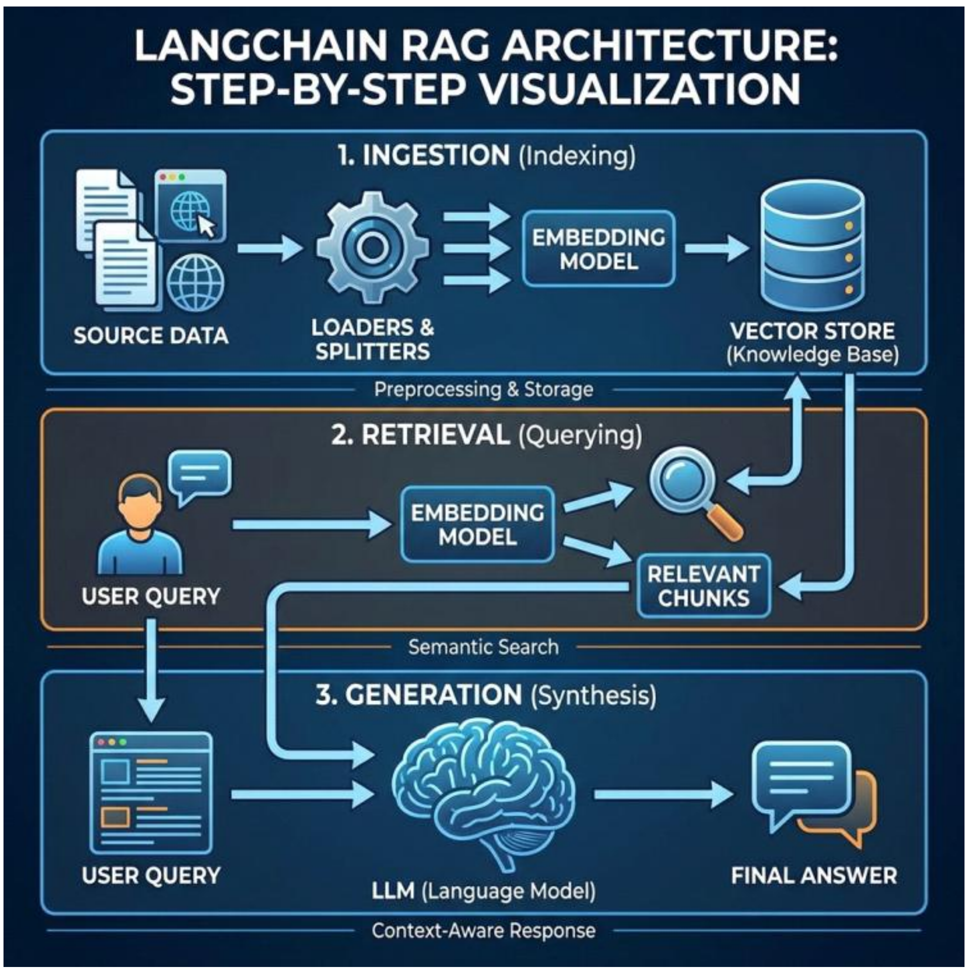
That "manager" is **LangChain**.

| Component | Purpose | Example |
|---|---|---|
| **Models** | Connects to LLMs and embedding models | GPT-4, Gemini, Claude, Llama |
| **Prompt Templates** | Creates reusable prompts with | "Explain {topic} in simple |
| **Chains (LCEL/Runnables)** | Connects multiple steps into a | Prompt → LLM → Output |
| **Agents** | Lets the LLM decide which tool to | ReAct Agent, Tool Calling Agent |
| **Tools** | External functions the agent can | Calculator, Search API, Python |
| **Retrievers** | Retrieves relevant documents | Vector Store Retriever |
| **Memory** | Maintains conversation history | Chat History, Conversation |
| **Document Loaders** | Loads data from different sources | PDF, DOCX, CSV, Websites |
| **Text Splitters** | Splits large documents into chunks | RecursiveCharacterTextSplitter |
| **Embeddings** | Converts text into vectors | OpenAI Embeddings, BGE, Hugging Face |
| **Vector Stores** | Stores and searches embeddings | FAISS, Chroma, Pinecone, |
| **Output Parsers** | Converts LLM output into structured formats | JSON, Pydantic Models |
| **Callbacks** | Logs and monitors execution | Token usage, latency tracking |
| **Chat Message History** | Stores previous chat messages | Redis, SQL, File |
| **Indexes (through retrievers/vector stores)** | Enables efficient document search | Vector Index |
| **RAG Pipeline** | Combines retrieval with | Retrieve → Prompt → LLM |
| **LangServe** | Deploys LangChain applications as | FastAPI-based serving |
| **LangSmith** | Debugging, tracing, and evaluation platform | Prompt debugging, execution |
| **Streaming** | Streams responses token by token | Real-time chatbot responses |
| **Multi-Agent Support** | Multiple agents collaborate | Research Agent + Coding Agent |

**Llama Index vs Lang Chain:**

| **Feature** | **Llama Index** | **Lang Chain** |
|---|---|---|
| Primary focus | Data ingestion and retrieval | End-to-end LLM application orchestration |
| Best at | RAG & Docs querying | Agents, workflows, tools & chains |
| Learning curve | Generally simpler | Broader, often more complex |
| Data connectors | Excellent | Good |
| Agent capabilities | Basic to moderate | Extensive |
| Typical use case | Search your documents | Build complete AI assistants and workflows |

They are not competitors so much as complementary tools. Many production systems use LlamaIndex for retrieval and LangChain for orchestration.




## Llama Index:-

1. Data Ingestion
2. Search
3. Retrieval
4. Indexing
5. Ranking and Structuring

**Points to be remember about Llama Index:**

- LlamaIndex is specifically designed for **search, retrieval, and question-answering applications**.
- It acts as a bridge between your data and Large Language Models (LLMs).
- It provides powerful **data ingestion** capabilities for integrating custom data with LLMs.
- It supports **structured, semi-structured, and unstructured** data sources.
- It automatically performs document parsing, chunking, embedding generation, indexing, retrieval, and response synthesis.
- It integrates with many vector databases such as **FAISS, Chroma, Pinecone, Milvus, Qdrant, and Weaviate**.
- It works with multiple embedding models and LLM providers, including OpenAI, Google Gemini, Anthropic Claude, Meta Llama, Ollama, Hugging Face, and others.
- It is one of the most widely used frameworks for building **Retrieval-Augmented Generation (RAG)** applications.

## Interview Definition:

Llama Index is an open-source data framework that connects LLM with private data by performing data **ingestion, Indexing, Semantic search, Retrieval, and Context preparation,** enabling accurate **Retrieval-Augmented Generation** (RAG) applications.

Use Llama Index when the primary challenge is connecting an LLM to your data through **ingestion, indexing, and retrieval**—for example, building a **RAG system** over **PDFs or databases.** Use Lang Chain when you need to orchestrate LLM workflows, such as **agents, tool calling, API integrations, memory, or multi-step automation.** In many production systems, they are used together: **Llama** Index handles retrieval, while **Lang Chain** manages the overall application workflow.

## Easy way to remember

| **Framework** | **Think of it as** | **Main Job** |
|---|---|---|
| Llama Index | Librarian | Finds the right information from your data |
| Lang Chain | Project Manager | Coordinates the entire workflow, tools, and decision-making |